

Day 08 – Cloud Server Setup: Docker, Nginx & Web Deployment

Step 1: Launched EC2 instance

Step 2: Connect via SSH

```
chmod 400 your-revision-key.pem  
ssh -i your-revision-key.pem ubuntu@<your-instance-ip>
```

Part 2: Install Docker & Nginx

Step 1: Update System

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install Docker

```
sudo apt install docker.io -y  
sudo systemctl start docker  
sudo systemctl enable docker
```

Verify:

```
docker --version  
systemctl status docker
```

Step 3: Install Nginx

```
sudo apt install nginx -y
```

Verify Nginx:

```
systemctl status nginx
```

You should see **active (running)** ✓

Part 3: Security Group Configuration

Ensure **Port 80 (HTTP)** is open in your cloud firewall.

Test Web Access

Open browser and visit:

`http://<your-instance-ip>`

✔ You should see **“Welcome to nginx!”**

Part 4: Extract Nginx Logs

Step 1: View Nginx Logs

```
sudo tail -n 20 /var/log/nginx/access.log
sudo tail -n 20 /var/log/nginx/error.log
```

Step 2: Save Logs to a File

```
sudo cat /var/log/nginx/access.log /var/log/nginx/error.log > ~/nginx-logs.txt
```

Verify:

```
ls -l nginx-logs.txt
head nginx-logs.txt
```

Step 3: Download Log File to Local Machine

AWS

```
scp -i your-key.pem ubuntu@<your-instance-ip>:~/nginx-logs.txt .
```

Commands Used

- `ssh -i your-key.pem ubuntu@<ip>`
- `sudo apt update && sudo apt upgrade -y`
- `sudo apt install docker.io -y`
- `sudo apt install nginx -y`
- `systemctl status nginx`
- `cat /var/log/nginx/access.log`
- `scp nginx-logs.txt`

Challenges Faced

- Initially could not access Nginx from browser
- Fixed by opening port 80 in security group

What I Learned

- How to launch and access a cloud server

- How to install and manage services using systemctl
- How to expose applications securely using security groups
- How to collect and download server logs

Why This Matters for DevOps

This task demonstrates real-world DevOps skills including cloud provisioning, remote server management, service deployment, log analysis, and security con